

Package ‘Ulysses’

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Title Automate OHDSI Study Setup

Version 1.1.0

Description

Automates setup of OHDSI study and provides functions to assist on improving organization and communication of a study.

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Imports cli,
crayon,
fs,
gert,
gh,
glue,
keyring,
lifecycle,
lubridate,
magrittr,
purrr,
rlang,
rstudioapi,
scales,
snakecase,
usethis,
whisker,
withr,
yaml,
RJSONIO

Suggests knitr,
rmarkdown,
here,
testthat (>= 3.0.0)

Config/testthat/edition 3

VignetteBuilder knitr

R topics documented:

addGitRemoteToUlysses	2
buildStudyHub	3
cloneUlysses	3
createExecutionSettings	4
defineLoadTable	5
execStudyPipeline	5
execStudyTask	6
ExecutionSettings	6
getAnalysisCohortsByCategory	8
importAtlasCohortsFromManifest	8
importAtlasConceptSetsFromManifest	9
initializeManifest	10
launchUlyssesRemoteWithBitBucketDC	10
launchUlyssesRepo	11
makeExecOptions	11
makeStudyMeta	12
makeTaskFile	12
makeUlyssesStudySettings	13
populateManifest	14
setAtlasConnection	14
setContributor	15
setDbConfigBlock	15
syncUlyssesWork	16
templateAtlasCredentials	16
updateStudyVersion	17
zipAndArchive	17
Index	18

addGitRemoteToUlysses *Function to add a Remote to Ulysses directory*

Description

This function adds a git remote to the Ulysses repo. If user adds a commit message it will add and commit files prior to adding and pushing to remote. This function will check to see if there are untracked files that need to be committed prior to adding remote.

Usage

```
addGitRemoteToUlysses(
  gitRemoteUrl,
  gitRemoteName = "origin",
  commitMessage = NULL
)
```

Arguments

gitRemoteUrl a character string of a git remote url
gitRemoteName The name of the remote, defaults to origin
commitMessage a character string of a commit Message to use. if null then skips commit

buildStudyHub	<i>Build Study Hub</i>
---------------	------------------------

Description

Build Study Hub

Usage

```
buildStudyHub(projectPath = here::here(), previewHub = TRUE)
```

Arguments

projectPath the path to the Ulysses repo, by default takes the path of the active R project
previewHub toggle to preview the hub after it builds. Default is TRUE

Value

invisible return. Creates _site folder with html files to preview site

cloneUlysses	<i>Function to clone a Ulysses Repo into local</i>
--------------	--

Description

Function to clone a Ulysses Repo into local

Usage

```
cloneUlysses(gitRemoteUrl, repoFolder)
```

Arguments

gitRemoteUrl the url of the git remote used to clone
repoFolder the location where you wish to place the git repo

Value

invisible return. Clones the git remote into local file structure and infuses default folders in clone

 createExecutionSettings

Create an ExecutionSettings object and set its attributes

Description

Create an ExecutionSettings object and set its attributes

Usage

```
createExecutionSettings(
    connectionDetails,
    connection = NULL,
    cdmDatabaseSchema,
    workDatabaseSchema,
    tempEmulationSchema,
    cohortTable,
    cdmSourceName
)
```

Arguments

connectionDetails	A DatabaseConnector connectionDetails object (optional if connection is specified)
connection	A DatabaseConnector connection object (optional if connectionDetails is specified)
cdmDatabaseSchema	The schema of the OMOP CDM database
workDatabaseSchema	The schema to which results will be written
tempEmulationSchema	Some database platforms like Oracle and Snowflake do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
cohortTable	The name of the table where the cohort(s) are stored
cdmSourceName	A human-readable name for the OMOP CDM source

Value

An ExecutionSettings object

defineLoadTable	<i>Define the load table</i>
-----------------	------------------------------

Description

Define the load table

Usage

```
defineLoadTable(atlasId, label, category, subCategory)
```

Arguments

label	a vector of naming labels to identify the atlas Ids, must be in order of atlas ids
category	a vector of categories to identify the atlas ids, must be in order of atlas ids
subCategory	a vector of sub-categories to identify the atlas ids, must be in order of atlas ids
atlas	a vector of atlas Ids to load

Value

a tibble where each row is an atlas asset with meta data

execStudyPipeline	<i>Function to execute all study task in analysis folder on set of config-Block</i>
-------------------	---

Description

Function to execute all study task in analysis folder on set of configBlock

Usage

```
execStudyPipeline(configBlock, env = rlang::caller_env())
```

Arguments

configBlock	name of one or multiple configBlock to use in the execution
env	the execution environment

execStudyTask	<i>Function to execute a study task in Ulysses</i>
---------------	--

Description

Function to execute a study task in Ulysses

Usage

```
execStudyTask(taskFile, configBlock, env = rlang::caller_env())
```

Arguments

taskFile	the name of the taskFile. Only use the base name
configBlock	the name of the configBlock to use in the execution
env	the execution environment

ExecutionSettings	<i>ExecutionSettings</i>
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Description

An R6 class to define an ExecutionSettings object

Active bindings

cdmDatabaseSchema	the schema containing the OMOP CDM
workDatabaseSchema	the schema containing the cohort table
tempEmulationSchema	the schema needed for temp tables
cohortTable	the table containing the cohorts
cdmSourceName	the name of the source data of the cdm

Methods

Public methods:

- [ExecutionSettings\\$new\(\)](#)
- [ExecutionSettings\\$getDbms\(\)](#)
- [ExecutionSettings\\$connect\(\)](#)
- [ExecutionSettings\\$disconnect\(\)](#)
- [ExecutionSettings\\$getConnection\(\)](#)
- [ExecutionSettings\\$clone\(\)](#)

Method new():

Usage:

```
ExecutionSettings$new(  
  connectionDetails = NULL,  
  connection = NULL,  
  cdmDatabaseSchema = NULL,  
  workDatabaseSchema = NULL,  
  tempEmulationSchema = NULL,  
  cohortTable = NULL,  
  cdmSourceName = NULL  
)
```

Arguments:

`connectionDetails` a connectionDetails object

`connection` a connection to a dbms

`cdmDatabaseSchema` The schema of the OMOP CDM database

`workDatabaseSchema` The schema to which results will be written

`tempEmulationSchema` Some database platforms like Oracle and Snowflake do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

`cohortTable` The name of the table where the cohort(s) are stored

`cdmSourceName` A human-readable name for the OMOP CDM source

Method `getDbms()`: extract the dbms dialect

Usage:

```
ExecutionSettings$getDbms()
```

Method `connect()`: connect to dbms

Usage:

```
ExecutionSettings$connect()
```

Method `disconnect()`: disconnect from dbms

Usage:

```
ExecutionSettings$disconnect()
```

Method `getConnection()`: retrieve the connection object

Usage:

```
ExecutionSettings$getConnection()
```

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
ExecutionSettings$clone(deep = FALSE)
```

Arguments:

`deep` Whether to make a deep clone.

```
getAnalysisCohortsByCategory
```

function to subset the cohortManifest based on category and sub category

Description

function to subset the cohortManifest based on category and sub category

Usage

```
getAnalysisCohortsByCategory(  
  cohortCategory,  
  cohortSubCategory = NULL,  
  cohortManifestPath = here::here("inputs/cohorts/cohortManifest.csv")  
)
```

Arguments

`cohortCategory` the category label in the cohort manifest
`cohortSubCategory`
the subCategory label to filter the cohort manifest, default is null
`cohortManifestPath`
the path to the manifest. Defaults to the standard ulysses path

Value

a tibble subsetting the cohort manifest by the id and label

```
importAtlasCohortsFromManifest
```

Import Atlas Cohorts from the manifest

Description

Import Atlas Cohorts from the manifest

Usage

```
importAtlasCohortsFromManifest(  
  cohortManifest,  
  atlasConnection,  
  outputFolder = here::here("inputs/cohorts/json")  
)
```


Arguments

atlasConnection an WebApiConnection R6 class that holds the creds to connect to webapi

outputFolder the location where the cohorts should be written

conceptSetManifest the location of the cohort manifest

Value

invisible return but stores the cohort files into Ulysses

importAtlasConceptSetsFromManifest
Import Atlas Concept sets from the manifest

Description

Import Atlas Concept sets from the manifest

Usage

```
importAtlasConceptSetsFromManifest(  
  conceptSetManifest,  
  atlasConnection,  
  outputFolder = here::here("inputs/conceptSets/json")  
)
```

Arguments

conceptSetManifest the location of the concept set manifest

atlasConnection an WebApiConnection R6 class that holds the creds to connect to webapi

outputFolder the location where the concept sets should be written

Value

invisible return but stores the concept set files into Ulysses

initializeManifest	<i>Initialize Manifests</i>
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Description

Initialize Manifests

Usage

```
initializeManifest(
  manifestType,
  loadTable = NULL,
  overwrite = FALSE,
  repoPath = here::here("inputs")
)
```

Arguments

manifestType	the type of manifest to initialize, either conceptSet or cohort
loadTable	a tibble made using defineLoadTable which prespecifies atlas ids to initialize the manifest
overwrite	toggle whether to overwrite existing manifests. Default is FALSE
repoPath	the location of the repo specifying where to save the manifest files

Value

invisible return but initializes the manifest in its appropriate folder location

launchUlyssesRemoteWithBitBucketDC	<i>Launch Ulysses Remote using Bitbucket Data Center</i>
------------------------------------	--

Description

Launch Ulysses Remote using Bitbucket Data Center

Usage

```
launchUlyssesRemoteWithBitBucketDC(repoName, hostUrl, httpToken, projectName)
```

Arguments

repoName	the name of the repository in project
hostUrl	the url hosting bitbucket data center
httpToken	the http access token configured to your bitbucket profile. To find the httpToken go to Manage Account > HTTP access tokens. It is recommended that you store the httpToken as an environment variable for ease of use.
projectName	the name of the project in Bitbucket Data Center storing the repo

Value

invisible return but initializes the remote on BitbucketDC

launchUlyssesRepo	<i>Function to Launch new Ulysses Repo</i>
-------------------	--

Description

Function to Launch new Ulysses Repo

Usage

```
launchUlyssesRepo(ulyssesStudySettings, verbose = TRUE, openProject = FALSE)
```

Arguments

ulyssesStudySettings	UlyssesStudy R6 class with the ulysses study details to make
verbose	a toggle whether to print details of launch in console
openProject	a toggle whether to open the repo as a new R project

Value

invisible return. Creates the ulysses repo in the local file structure

makeExecOptions	<i>Make ExecOptions for Ulysses</i>
-----------------	-------------------------------------

Description

Make ExecOptions for Ulysses

Usage

```
makeExecOptions(
  dbms,
  workDatabaseSchema,
  tempEmulationSchema = NULL,
  dbConnectionBlocks
)
```

Arguments

dbms	specify the dbms used in the exec options
workDatabaseSchema	the name of the workDatabaseSchema as a character string, location in DB where user has write access
tempEmulationSchema	the name of the tempEmulationSchema as a character strings
dbConnectionBlocks	a list of DbConfigBlock R6 classes specifying the dbs to connect

Value

A ExecOptions R6 class with the execOptions

makeStudyMeta	<i>Make Study Meta for Ulysses</i>
---------------	------------------------------------

Description

Make Study Meta for Ulysses

Usage

```
makeStudyMeta(
  studyTitle,
  therapeuticArea,
  studyType,
  contributors,
  studyLinks = NULL,
  studyTags = NULL
)
```

Arguments

studyTitle	the title of the study as a character string
therapeuticArea	the TA as a character string
studyType	the study type (typically characterization)
studyLinks	a list of study links
studyTags	a list of study tags

Value

A StudyMeta R6 class with the study meta

makeTaskFile	<i>Function initializing an R file for an analysis task</i>
--------------	---

Description

Function initializing an R file for an analysis task

Usage

```
makeTaskFile(
  nameOfTask,
  author = NULL,
  description = NULL,
  projectPath = here::here(),
  openFile = TRUE
)
```

Arguments

nameOfTask	The name of the analysis task script
author	the name of the person authoring the file. Defaults to template text if NULL
description	a description of the analysis task. Defaults to template text if NULL
projectPath	the path to the project
openFile	toggle on whether the file should be opened

`makeUlyssesStudySettings`*Make Ulysses Study Settings*

Description

Make Ulysses Study Settings

Usage

```
makeUlyssesStudySettings(  
  repoName,  
  repoFolder,  
  studyMeta,  
  execOptions,  
  gitRemote = NULL,  
  renvLock = NULL  
)
```

Arguments

repoName	the name of repo as a character string
repoFolder	the folder path where the repo is stored in local as a character string
studyMeta	a StudyMeta R6 class with the details describing the study
execOptions	a ExecOptions R6 class with the execution details needed for the study
gitRemote	a remote url used to clone and set remote git
renvLock	file path to a renvLock file

Value

A UlyssesStudy R6 class with the ulysses study details to make

populateManifest	<i>Populate Manifests</i>
------------------	---------------------------

Description

Populate Manifests

Usage

```
populateManifest(
  manifestType,
  importFromAtlas = FALSE,
  atlasConnection = setAtlasConnection(),
  repoPath = here::here("inputs")
)
```

Arguments

manifestType	the type of manifest to initialize, either conceptSet or cohort
importFromAtlas	toggle whether to import content from atlas. Default is TRUE
atlasConnection	add atlasConnection object.
repoPath	the location of the repo specifying where to save the manifest files

Value

invisible return but populates the manifest in its appropriate folder location

setAtlasConnection	<i>Set Atlas Connection</i>
--------------------	-----------------------------

Description

Set Atlas Connection

Usage

```
setAtlasConnection()
```

Value

an R6 class of WebApiConnection

setContributor	<i>Set Ulysses Contributor</i>
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Description

Set Ulysses Contributor

Usage

```
setContributor(name, email, role)
```

Arguments

name	the name of the contributor as a character string
email	the email of the contributor as a character string
role	the role of the contributor as a character string

Value

A ContributorLine R6 class with the contributor info

setDbConfigBlock	<i>set the config block for a database</i>
------------------	--

Description

set the config block for a database

Usage

```
setDbConfigBlock(
  configBlockName,
  cdmDatabaseSchema,
  cohortTable,
  databaseName = NULL,
  databaseLabel = NULL
)
```

Arguments

configBlockName	the name of the config block
cdmDatabaseSchema	the cdmDatabaseSchema specified as a character string
cohortTable	a character string specifying the way you want to name your cohort table
databaseName	the name of the database, typically uses the db name and id. For example optum_dod_202501
databaseLabel	the labelling name of the database, typically a common name for a db. For example Optum DOD

Value

A StudyMeta R6 class with the study meta

syncUlyssesWork	<i>Function to sync local work in Ulysses to remote</i>
-----------------	---

Description

Function to sync local work in Ulysses to remote

Usage

```
syncUlyssesWork(  
  commitMessage,  
  branch = gert::git_branch(),  
  gitRemoteName = "origin"  
)
```

Arguments

commitMessage	the commit message describing the work done since the last commit
branch	the name of the branch you want to commit changes. Defaults to the active branch
gitRemoteName	the name of the remote. Defaults to origin.

templateAtlasCredentials	<i>Template for setting Atlas Credentials</i>
--------------------------	---

Description

Template for setting Atlas Credentials

Usage

```
templateAtlasCredentials()
```

Value

no return; prints info to console

updateStudyVersion	<i>Function to update the study version</i>
--------------------	---

Description

Function to update the study version

Usage

```
updateStudyVersion(versionNumber, projectPath = here::here())
```

Arguments

versionNumber	the semantive version number to set as the new project version: 1.0.0
projectPath	the path of the project, defaults to the directory of the active Ulysses project

zipAndArchive	<i>Zip and Archive results from a study execution</i>
---------------	---

Description

Zip and Archive results from a study execution

Usage

```
zipAndArchive(input)
```

Arguments

input	the type of files to zip and archive. There are three options exportMerge, exportPretty and site. exportMerge is the merged results in long format. The exportPretty are xlsx files with formatted output from the study. The site is the html files of the studyHub
-------	--

Value

invisible return. Stores the input as a zip file in the exec/archive folder

Index

`addGitRemoteToUlysses`, [2](#)

`buildStudyHub`, [3](#)

`cloneUlysses`, [3](#)

`createExecutionSettings`, [4](#)

`defineLoadTable`, [5](#)

`execStudyPipeline`, [5](#)

`execStudyTask`, [6](#)

`ExecutionSettings`, [6](#)

`getAnalysisCohortsByCategory`, [8](#)

`importAtlasCohortsFromManifest`, [8](#)

`importAtlasConceptSetsFromManifest`, [9](#)

`initializeManifest`, [10](#)

`launchUlyssesRemoteWithBitBucketDC`, [10](#)

`launchUlyssesRepo`, [11](#)

`makeExecOptions`, [11](#)

`makeStudyMeta`, [12](#)

`makeTaskFile`, [12](#)

`makeUlyssesStudySettings`, [13](#)

`populateManifest`, [14](#)

`setAtlasConnection`, [14](#)

`setContributor`, [15](#)

`setDbConfigBlock`, [15](#)

`syncUlyssesWork`, [16](#)

`templateAtlasCredentials`, [16](#)

`updateStudyVersion`, [17](#)

`zipAndArchive`, [17](#)